Forces in Mechanics xxx (xxxx) 100352

Contents lists available at ScienceDirect

Forces in Mechanics

journal homepage: www.elsevier.com

Journal Cover Image

Insight of thermal transport in Maxwell fluid flow on a porous contracting flat surface inside a porous material with non-uniform heat source/sink: Multiple solutions and stability treatment

Amit Kumar Pandey [a], Prakash Goswami [b], Sohita Rajput [c, *], Krishnendu Bhattacharyya [b, *], Ali J. Chamkha [d], Dhananjay Yadav [e]

[a] Department of Mathematics, School of Basic Sciences, Chandigarh University, Unnao, Lucknow-209859, Uttar Pradesh, India
[b] Department of Mathematics, Institute of Science, Banaras Hindu University, Varanasi-221005, Uttar Pradesh, India
[c] Mathematics Section, Department of Science and Humanity, Sershah Engineering College, Sasaram-821113, Bihar, India
[d] Faculty of Engineering, Kuwait College of Science and Technology, Doha District, Kuwait
[e] Department of Mathematical and Physical Sciences, University of Nizwa, Oman

ARTICLE INFO

Keywords:
Heat transfer
Maxwell fluid flow
Contracting flat surface
Non-uniform heat source/sink
Dual solutions and stability analysis

ABSTRACT

The thermal behaviour of viscoelastic fluids, particularly those described by the Maxwell model capable of capturing stress-relaxation effects, exhibits several intriguing and practically relevant characteristics. Likewise, flow induced by a contracting surface presents distinctive and non-classical boundary layer features. Motivated by these aspects, the present theoretical study investigates heat transfer in Maxwell fluid flow over a porous contracting flat surface embedded in a porous medium, incorporating a temperature-dependent and spatially-dependent non-uniform heat source/sink. The governing partial differential equations are reduced to self-similar ordinary differential equations using appropriate similarity transformations, and the resulting nonlinear system is solved with high accuracy using the shooting method along with fourth-order Runge–Kutta (RK-4) technique and Secant method. For sufficiently strong suction, dual solutions of the transformed equations are obtained, and a stability assessment confirms that the upper branch solutions are stable, while the lower branch solutions are unstable. The comprehensive analysis uncovers several notable physical insights. The analysis shows that viscoelasticity of Maxwell fluids plays a key stabilizing role: increasing the Deborah number weakens vorticity production, lowers the suction required to maintain an attached boundary layer, and broadens the parameter range in which similarity solutions exist. Permeability of the porous medium also strongly influences boundary layer behaviour; higher resistance suppresses vorticity generated by sheet contraction, enhancing skin friction and heat transfer in the upper branch but diminishing both in the lower branch. Both temperature- and space-dependent heat sources reduce heat transfer rates, while their sink counterparts enhance cooling, though spatial heat generation/absorption produces a far more pronounced thermal response. Suction strengthens and compresses the boundary layer in the stable upper branch but weakens the flow in the unstable lower branch. Overall, the study clarifies how elasticity, permeability, heat generation mechanisms, and mass transfer collectively shape the dual solution structure of Maxwell fluid boundary layers, offering detailed insights relevant to thermal control in viscoelastic–porous systems.

Nomenclature

A^*	Coefficient associated with temperature-dependent source/sink of heat
B^*	Coefficient associated with spatially-dependent source/sink of heat
c	Contraction constant
C_f	Local skin-friction
c_p	Specific heat
$f(\eta)$	A dimensionless function

* Corresponding authors.
E-mail addresses: sohita.math@gmail.com (S. Rajput), krish.math@yahoo.com (K. Bhattacharyya).

https://doi.org/10.1016/j.finmec.2026.100352
2666-3597/© 20XX

UNCORRECTED PROOF

Note: Low-resolution images were used to create this PDF. The original images will be used in the final composition.

k	Porous-medium permeability
K	Permeability parameter
Nu	Local Nusselt number
Pr	Prandtl number
q_w	wall heat flux
Re_x	Local Reynolds number
S	Surface mass transfer parameter
T	Temperature
T_w	Constant surface temperature
T_∞	Fixed free stream temperature faraway from surface
u	Velocity component along x-axis
U_w	Contracting surface velocity
v	Velocity component along y-axis
v_w	Velocity of wall mass transfer
z_1, z_2, z_3, z_4, z_5	Variables

Greek symbols

β	Deborah number/Maxwell fluid parameter
$\Delta\eta$	Step size of computation
η	Similarity variable
η_∞	Suitable finite value of infinity condition of η
$\theta(\eta)$	A dimensionless function
κ	Thermal conductivity
λ	Maxwell relaxation time
μ	Dynamic viscosity
υ	Kinematic viscosity
ρ	Density
ρc_p	Heat capacity
τ_w	Wall shear stress
ψ	Stream function

1. Introduction

In the realm of fluid flow research, heat transfer plays a crucial role and has been extensively studied for decades to improve the quality of the products manufactured across various industries sectors. Applications of heat transport are abundant in heat exchangers, nuclear reactor cooling, condensers, and thermal regulation of storage tanks, among others [1]. It is well-known that if there is a temperature difference between two regions, then the heat transfer phenomenon is observed from higher towards lower temperature regions. When a fluid flows past a heated or cooled surface, a thermal boundary layer region develops adjacent to solid surface due to the exchange of thermal energy between neighboring fluid layers. Governed by the energy conservation equation, this boundary layer is fundamental in understanding convective thermal behaviour in fluid flows. Conduction and convection are two very frequently occurring mechanisms for the transportation of heat in fluids, between solid surface and fluid, and within the fluid itself. Several factors including surface temperature, ambient fluid conditions, thermophysical characteristics, surface roughness and its type, and fluid type, significantly influence heat transfer rates. Acrivos [2] analyzed heat transfer within the boundary layer under combined forced and free convection, accounting for gravitational coupling between momentum and energy equations. Further developments include the study of convective heat transfer on a flat plate with pressure gradient by Fathizadeh and Rashidi [3]. Roşca et al. [4] and Parida et al. [5] conducted heat transfer flow over a uniformly moving flat plate under partial slip conditions. In particular, Parida et al. [5] investigated radiative heat transfer with variable fluid properties.

While the aforementioned works primarily examine Newtonian fluids, many industrial processes employ non-Newtonian fluids due to their unique rheological characteristics. Non-Newtonian models are governed by nonlinear stress-strain relationships. Research on the heat exchange behaviour of non-Newtonian fluids is increasingly relevant due to its variety of applications, such as, polymer processing, slurry fuels, colloidal and suspension solutions, and exotic lubricants. Several studies have addressed heat transport in non-Newtonian fluids. For instance, Sohail et al. [6] analyzed radiative transport in Sutterby nanofluids, Yusuf et al. [7] examined magneto-bioconvective Williamson nanofluid flow, and Qaiser et al. [8] explored Walters-B nanofluid subjected to Newtonian heating conditions. Shatnawi et al. [9] conducted a comprehensive investigation of the unsteady stagnation-point flow of Casson hybrid nanofluids, incorporating the Lorentz force and employing three different nanofluid thermal conductivity models.

Non-Newtonian fluids are classified into various subcategories based on their rheological behaviour, among which Maxwell fluid represents a distinct viscoelastic type exhibiting both viscous and elastic characteristics. Among various non-Newtonian fluid models, the Maxwell viscoelastic model has received considerable interest because of its ability to describe stress relaxation phenomena. According to Maxwell [10], the elastic nature of a fluid is associated with energy storage, whereas its viscous behaviour is linked to energy dissipation. For investigating polymeric fluids, the Maxwell model [10] serves as a simple and practical framework for analyzing stress relaxation effects. Several notable findings related to Maxwell fluid flow had been reported by Vieru et al. [11] and Hayat et al. [12]. Olsson et al. [13] contributed to the understanding of various characteristics of viscoelastic fluid flow. In a related study, Choi et al. [14] investigated Maxwell fluid flow in a channel with suction. Exact solutions to Maxwell fluid flow configurations were obtained by Fetecau and Fetecau [15,16]. Additional important developments in Maxwell fluid models had been reported by Mukhopadhyay and Bhattacharyya [17], Zhang et al. [18] and Akyildiz and Alshammari [19].

The theoretical investigation of heat transfer in flows induced by a contracting sheet has recently gained attention, as it leads to distinct thermodynamic behaviour compared to expanding sheet flows. Unlike the relatively stable flow generated by an expanding surface, contraction produces a more complex flow field, where reverse or backward motion induces vorticity and disrupts the flow structure. To preserve the boundary layer formation under such conditions, additional counteracting forces are required to confine the generated vorticities within the boundary layer region. Building on this line of research, Miklavčič and Wang [20] showed that viscous flow over a contracting surface with mass suction admits a self-similar boundary-layer solution that is not unique, and they also derived an exact analytical form. Subsequent studies expanded this framework: Sajid and Hayat [21] and Hayat et al. [22] developed semi-analytical solutions for viscous and second-grade fluid models, respectively, while Sajid et al. [23] investigated MHD rotational flow of Newtonian fluid past a contracting sheet. Studies by Ishak et al. [24] and Turkyilmazoglu [25] examined the flow of power-law and viscoelastic fluids over a contracting surface, respectively. Moreover, Wang [26] proposed that applying a stagnation flow toward the contracting sheet can effectively suppress or retard boundary-layer separation. Further developments on contracting sheet flow were made by Ishak et al. [27], Bhattacharyya et al. [28], Bhattacharyya and Layek [29], and Ismail et al. [30] by introducing additional physical effects. Later, researchers [31–40] explored heat transfer characteristics in Newtonian and non-Newtonian flows over contracting and expanding surfaces with diverse boundary conditions and flow limitations. It is well established that the surface temperature boundary condition is a key factor governing the rate of heat transfer.

Investigating flow within porous materials is of significant interest, as the solid matrix with void spaces enables regulation of fluid motion

UNCORRECTED PROOF

and thermal transport within the boundary layer, making it suitable for enhanced cooling applications. In this context, porosity serves as a critical parameter influencing heat transfer behaviour. Porosity is defined as the ratio of void volume to total volume of a material. For low-velocity flow in low-porosity media, Darcy's empirical law states that the pressure drop varies linearly with flow velocity. Common natural and artificial porous media include sand, pumice, dolomite, sponge, foamed plastics, and even biological systems like the human lung. Flows through porous media are widely encountered in engineering and industrial practice, with applications ranging from filtration and purification, pollutant migration in soils and aquifers, and drying of porous substances to food preservation, radioactive waste disposal, and thermal management of nuclear fuel debris. Research on porous medium flow has progressed through notable contributions. Vafai and Tien [41] examined the effects of inertia and solid boundaries on heat and fluid transport within porous structures. Jayanthi and Kumari [42] presented results on non-Newtonian mixed convection along a vertical surface embedded in a porous medium, while Aziz et al. [43] focused on power-law fluid behaviour with suction and blowing in Darcy-type porous material. Shatnawi et al. [44] further advanced the field by comparing flow of Casson hybrid nanofluid using the Yamada–Ota, Xue, and Tiwari–Das thermal conductivity models subject to induced magnetic field within a porous matrix. More recent publications [45–47] continue to address porous medium flows across a range of fluid and boundary conditions.

The incorporation of heat sinks or sources plays a crucial role in heat transfer analyses, with applications in areas such as nuclear reactor cooling, semiconductor manufacturing, and electronic component thermal management. These phenomena are used to regulate or enhance heat transfer performance effectively. Although temperature-dependent heat source/sink models are common in the literature, more accurate predictions have been achieved in recent studies by considering models dependent on both temperature and spatial variables. Abd El-Aziz [48] studied stagnation flow and heat transfer over expanding and contracting surfaces under non-uniform heat source/sink and heat flux conditions. Mythili and Sivaraj [49] examined unsteady Casson fluid flow over a vertical cone and flat plate considering variable fluid properties, higher-order chemical reactions, and both surface- and temperature-dependent heat source/sink. Ramandevi et al. [50] extended this work by incorporating viscous dissipation, MHD effects, and the Cattaneo–Christov heat flux model. Further advancements include the work of Alzahrani and Khan [51], who investigated the impact of non-uniform heat source/sink and activation energy on three-dimensional flow through Darcy–Forchheimer porous media. Manvi et al. [52] extended this line of research to the Eyring–Powell fluid model over an expanding sheet under non-uniform heat generation or absorption. In addition, Abbas et al. [53] studied power-law nanofluid flow past a vertical Riga surface while accounting for heat generation and variations in viscosity and thermal conductivity.

1.1. Problem statement

The present work considers the steady two-dimensional flow and heat transfer of a Maxwell viscoelastic fluid driven by a continuously contracting sheet embedded in a Darcy-type porous medium under the influence of mass suction. A generalized non-uniform heat source/sink model dependent on both spatial coordinates and fluid temperature is incorporated to represent realistic thermal environments. The governing nonlinear partial differential equations describing momentum and energy transport are transformed into similarity-based ordinary differential equations and solved numerically using the shooting method combined with the Runge–Kutta fourth-order scheme. Later, the impacts key dimensionless parameters on the flow structure, thermal boundary layer thickness, surface shear stress, and heat transfer rate are systematically analyzed. The outcomes provide new insights into controlling flow stability and thermal performance in applications involving viscoelastic materials interacting with contracting geometries and porous structures.

1.2. Novelty of the study

The present investigation advances existing knowledge by introducing several novel features that distinguish this work from earlier studies. First, unlike classical Newtonian, the current research focuses on a Maxwell viscoelastic fluid, capable of capturing both relaxation behaviour and memory effects. This provides a more realistic representation for polymeric and industrial fluids where stress relaxation significantly influences transport phenomena. Second, the present work examines the fluid motion over a continuously contracting sheet, a configuration known to generate reverse flow and vorticity effects. Here, the contracting surface is combined with mass suction, which plays a crucial role in suppressing vorticity and maintaining boundary-layer structure—an aspect rarely addressed in earlier Maxwell fluid-based heat transfer studies. Third, a major novelty lies in the adoption of a combined space- and temperature-dependent non-uniform heat source/sink model, rather than the conventional constant or solely temperature-dependent models. This hybrid formulation offers a more generalized representation of heat generation or absorption mechanisms in realistic thermal systems, where heat gradients are neither uniform nor purely temperature-driven. Fourth, the current work incorporates the flow inside a Darcy porous medium, enabling the exploration of fluid–solid interactions and dissipation effects within a porous structure. The simultaneous consideration of viscoelasticity, contraction dynamics, porous permeability, and non-uniform thermal source/sink effects distinguishes this analysis from previous studies, which typically focused on these mechanisms in isolation.

2. Mathematical formulation

Take a two-dimensional viscous flow of a Maxwell fluid with heat transfer driven by a contracting surface whose velocity is U_w (= cx, with c the contraction constant). Using the boundary-layer assumptions, the governing conservation equations for mass, momentum and energy are expressed as follows [17,48]:

$$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}=0, \tag{1}$$

$$u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}+\lambda\left[u^2\frac{\partial^2 u}{\partial x^2}+v^2\frac{\partial^2 u}{\partial y^2}+2uv\frac{\partial^2 u}{\partial x\partial y}\right]=\upsilon\frac{\partial^2 u}{\partial y^2}-\frac{\upsilon}{k}u, \tag{2}$$

$$u\frac{\partial T}{\partial x}+v\frac{\partial T}{\partial y}=\frac{\kappa}{\left(\rho c_p\right)}\frac{\partial^2 T}{\partial y^2}+\frac{1}{\left(\rho c_p\right)}\overline{q}. \tag{3}$$

The coordinate system is chosen with the x-axis tangential to the surface and the y-axis normal to it; u and v denote the velocity components along x and y, respectively. The symbols λ, k, and T stand for the Maxwell relaxation time, the porous-medium permeability, and the temperature, respectively. The fluid properties appearing in the governing equations are μ (dynamic viscosity), ρ (density), $\upsilon=\mu/\rho$ (kinematic viscosity), ρc_p (heat capacity), and κ (thermal conductivity). In Eq. (2) the final term on the left-hand side is due to the Maxwell constitutive relation (non-Newtonian effect), whereas the terms on the right-hand side account for viscous stresses and flow resistance in the porous medium. In Eq. (3) the convective transport appears on the left, while thermal conduction and the heat source/sink appear on the right. The model geometry is illustrated in Fig. 1.

Non-uniform heat source/sink has identified as [49]:

$$\overline{q}=\frac{\kappa U_w}{\upsilon x}\left[A^*\left(T-T_\infty\right)+B^*\left(T_w-T_\infty\right)e^{-\frac{y}{x}\sqrt{\frac{U_w x}{\upsilon}}}\right], \tag{4}$$

UNCORRECTED PROOF

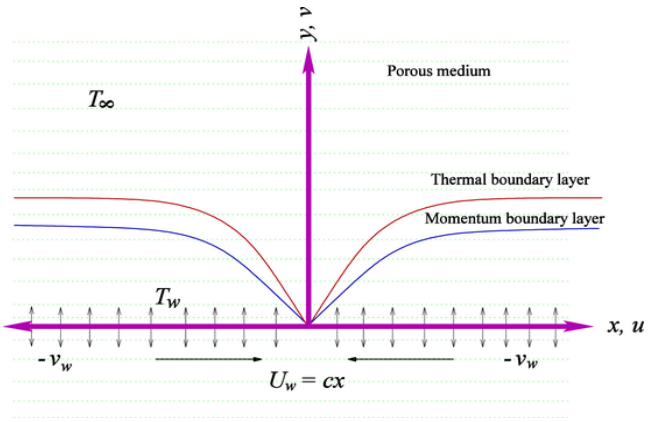


Fig. 1. Schematic representation of the physical model.

where A^* and B^* denote the coefficients associated with the temperature-dependent and spatially-dependent source/sink of heat, respectively. Here, it should be noted that $A^* > 0$ and $B^* > 0$ correspond heat source and $A^* < 0$ and $B^* < 0$ represent heat sink.

Hence using (4), the Eq. (3) reduces to:

$$u\frac{\partial T}{\partial x} + v\frac{\partial T}{\partial y} = \frac{\kappa}{(\rho c_p)}\frac{\partial^2 T}{\partial y^2} + \frac{\kappa}{(\rho c_p)}\frac{U_w}{\upsilon x}\left[A^*\left(T - T_\infty\right) + B^*\left(T_w - T_\infty\right)e^{-\frac{y}{x}\sqrt{\frac{U_w x}{\upsilon}}}\right]. \tag{5}$$

The relevant boundary conditions(BCs.) with mass transfer are:

$$u = -U_w = -cx,\ v = -v_w,\ T = T_w \text{ at } y = 0, \tag{6}$$

$$u \to 0,\ T = T_\infty \text{ as } y \to \infty, \tag{7}$$

where T_w denotes the constant surface temperature, T_∞ represents the fixed free stream temperature faraway from surface, with $T_w > T_\infty$. The quantity v_w indicates the velocity of wall mass transfer with $v_w > 0$ for suction and $v_w < 0$ for injection.

The Eqs. (1), (2) and (5) are transformed by employing the well-known stream function $\psi = \psi(x, y)$ obeying $u = \frac{\partial \psi}{\partial y}$ and $v = -\frac{\partial \psi}{\partial x}$. Under this transformation, Eq. (1) satisfies identically and Eqs. (2) and (5) become

$$\frac{\partial \psi}{\partial y}\frac{\partial^2 \psi}{\partial x \partial y} - \frac{\partial \psi}{\partial x}\frac{\partial^2 \psi}{\partial y^2} + \lambda\left[\left(\frac{\partial \psi}{\partial y}\right)^2\frac{\partial^3 \psi}{\partial x^2 \partial y} + \left(\frac{\partial \psi}{\partial x}\right)^2\frac{\partial^3 \psi}{\partial y^3} - 2\frac{\partial \psi}{\partial y}\frac{\partial \psi}{\partial x}\frac{\partial^3 \psi}{\partial x \partial y^2}\right] = \upsilon\frac{\partial^3 \psi}{\partial y^3} - \frac{\upsilon}{k}\frac{\partial \psi}{\partial y}, \tag{8}$$

$$\frac{\partial \psi}{\partial y}\frac{\partial T}{\partial x} - \frac{\partial \psi}{\partial x}\frac{\partial T}{\partial y} = \frac{\kappa}{(\rho c_p)}\frac{\partial^2 T}{\partial y^2} + \frac{\kappa}{(\rho c_p)}\frac{U_w}{\upsilon x}\left[A^*\left(T - T_\infty\right) + B^*\left(T_w - T_\infty\right)e^{-\frac{y}{x}\sqrt{\frac{U_w x}{\upsilon}}}\right], \tag{9}$$

respectively. Boundary conditions (6) and (7) are accordingly transformed to their reduced forms:

$$\frac{\partial \psi}{\partial y} = -cx,\ \frac{\partial \psi}{\partial x} = -v_w,\ T = T_w \text{ at } y = 0 \tag{10}$$

$$\frac{\partial \psi}{\partial y} \to 0,\ T \to T_\infty \text{ at } y \to \infty \tag{11}$$

To express ψ and T in non-dimensional form, following variables are introduced:

$$\psi = \sqrt{\upsilon x U_w} f(\eta) \text{ and } T = T_\infty + \left(T_w - T_\infty\right)\theta(\eta), \tag{12}$$

where similarity variable η is defined as $\eta = y\sqrt{\frac{U_w}{x\upsilon}}$.

Upon substituting these transformations into the governing equations, the following coupled nonlinear self-similar ODEs are achieved:

$$f''' + \beta\left(2fff'' - f^2 f''\right) + ff'' - f'^2 - Kf' = 0, \tag{13}$$

$$\theta'' + \mathrm{Pr} f\theta' + A^*\theta + B^* e^{-\eta} = 0 \tag{14}$$

with

$$\left.\begin{aligned} &f(0) = S, f'(0) = -1,\ f'(\infty) = 0, \\ &\theta(0) = 1 \text{ and } \theta(\infty) = 0. \end{aligned}\right\} \tag{15}$$

where $\beta = \lambda c$ denotes the Deborah number (or Maxwell fluid parameter), $K = \upsilon/(ck)$ indicates permeability parameter, $S = v_w/\sqrt{\upsilon c}$ is surface mass transfer parameter with $S > 0$ and < 0 corresponding to mass suction and injection, and $\mathrm{Pr} = \upsilon\rho c_p/\kappa$ states the Prandtl number.

Quantity of industrial significance associated with drag-force is the coefficient of local skin-friction, C_f directly linked to wall shear stress τ_w. It is defined as $C_f = \frac{\tau_w}{\rho U_w^2}$, where $\tau_w = \mu(1+\beta)\left(\frac{\partial u}{\partial y}\right)_{y=0}$, leading to the simplified expression $C_f \mathrm{Re}_x^{1/2} = (1+\beta) f''(0)$, where $\mathrm{Re}_x = U_w x/\upsilon$ gives local Reynolds number.

From an engineering standpoint, the physical quantity of interest related to the heat transfer rate is characterized by local Nusselt number, Nu, expressed as $Nu = \frac{x q_w}{\kappa(T_w - T_\infty)}$, where $q_w = -\kappa\left(\frac{\partial T}{\partial y}\right)_{y=0}$ is wall heat flux. This formulation yields $Nu\mathrm{Re}_x^{-1/2} = -\theta'(0)$.

3. Solution methodology

The two-point boundary value problem(BVP) arising from the nonlinear ordinary differential Eqs. (13) and (14) with (15) is solved by shooting method [54,55] by transforming the problem into an initial value problem(IVP). In this approach, it requires to select an suitable finite value of $\eta(\to \infty)$, say, η_∞. The corresponding first-order system for the IVP is then formulated as follows:

$$z'_1 = z_2,\ z'_2 = z_3,\ z'_3 = \left(z_2^2 - z_1 z_3 - 2\beta z_1 z_2 z_3 + k z_2\right)/\left(1 - \beta z_1^2\right) \tag{16}$$

$$z'_4 = z_5,\ z'_5 = -\mathrm{Pr} z_1 z_5 + A^* z_4 + B^* e^{-\eta}, \tag{17}$$

with

$$z_1(0) = S,\ z_2(0) = -1,\ z_4(0) = 1, \tag{18}$$

where $f(\eta) = z_1(\eta)$ and $\theta(\eta) = z_4(\eta)$.

Quantities $z_3(0)[f''(0)]$ and $z_5(0)[\theta'(0)]$ are required in order to solve Eqs. (16) and (17) together with the boundary conditions in (18) as an IVP, but these are not specified and must therefore be estimated. The fourth-order Runge–Kutta (RK-4) technique is employed to find numerical solution once initial guesses for $z_3(0) = f''(0)$ and $z_5(0) = \theta'(0)$ being assigned. Using the Secant method, these guesses are iteratively updated by comparing the computed values of $f'(\eta_\infty)$ and $\theta(\eta_\infty)$ $(\eta_\infty = 10)$ with specified BCs. The repeated application of the RK-4 and Secant methods yields progressively improved estimates. During computations, the step size is assumed to be $\Delta\eta = 0.01$. The iteration continues until the necessary precision of 10^{-5} with asymptotic convergence is achieved.

UNCORRECTED PROOF

4. Results and discussion

The numerical framework outlined above is utilized to investigate the existence, non-existence, uniqueness, and non-uniqueness of the two-dimensional steady Maxwell fluid flow over a porous contracting surface inside porous material with non-uniform heat source/sink. For the purpose of demonstrating the findings of investigated problem, fixed baseline values of the governing parameters are adopted whenever the parameters are not subjected to variation. These baseline values are taken as $\beta = 0.1$, $K = 0.1$, $S = 1.95$, $\mathrm{Pr} = 2$, $A^* = 0.1$, $B^* = 0.1$. To effectively demonstrate the parametric influences, the varying values of the parameters are selected such that the resulting trends in the velocity and temperature profiles are clearly visible while remaining within physically realistic and meaningful ranges.

4.1. Impacts on boundary layer existence, skin-friction and nusselt number

In the present flow configuration, the generation of vorticity induced by the contracting surface is reduced when the fluid type changes from Newtonian ($\beta = 0$) to Maxwell ($\beta > 0$). Consequently, the necessity of amount of mass suction via porous sheet to confine the vorticity within the boundary layer becomes smaller. As a result, the essential quantity of mass suction for a Maxwell fluid flow is less than that of a Newtonian fluid flow. Since the Deborah number (β) characterizes the fluid's intrinsic fluidity, where larger β corresponds to reduced fluidity and a positive β implies that the vorticity generated by surface contraction is weaker for a Maxwell fluid compared to a Newtonian fluid. Physically, a positive and increasing Deborah number means that the fluid retains memory of its past configuration for a longer duration, and reduced tendency for boundary layer separation. The viscoelastic relaxation inherent to Maxwell fluids allows part of the imposed deformation to be temporarily stored as elastic energy rather than immediately converted into vorticity-generating viscous shear. For instance, the minimum mass suction needed for Deborah number $\beta = 0.1$ is 1.8391 in order to preserve structure of boundary layer. Since separation of boundary layer for the Maxwell fluid ($\beta > 0$) is deferred for larger values of the Deborah number β, i.e., $\beta = 0.2$, similarity solutions are attainable with even smaller levels of mass suction, i.e., $S \geq 1.76205$. Thus, the interplay between viscoelastic relaxation and wall contraction reveals that elasticity acts as a stabilizing mechanism, reducing the suction requirement and broadening the range of conditions under which self-similar boundary layer solutions persist. Fig. 2 presents the variation of the skin-friction coefficient with the mass transfer parameter S for different values of β. The figure clearly demonstrates that for $\beta = 0.1$, dual similarity solutions exist when $S \geq 1.8391$ and no solution appears for $S < 1.8391$, indicating that separation of boundary layer happens. Additionally, for $\beta = 0.2$, when $S \geq 1.76205$ the dual solutions are obtained Maxwell fluid flow and boundary layer disappears for $S < 1.76205$(boundary layer separation). Moreover, it can be also noticed that skin-friction rises with augmentation of β in upper branch and it reduces for the lower branch. The variation in Nusselt number with S for several β values is displayed in Fig. 3. It has been witnessed that heat transport coefficient upsurges with the augment of β on upper branch solution and it drops for lower branch solution. Furthermore, it may be inferred that Maxwell non-Newtonian fluids cool more effectively than Newtonian fluids and so for real cooling purpose Maxwell fluids are more useful than the Newtonian fluids, making them advantageous in industrial cooling applications. Fig. 4 illustrates the variation of the skin-friction coefficient with S for different permeability parameters K. Skin-friction coefficient rises as the value of K grows in upper branch, while the opposite trend is noted on the lower branch. This distinction arises from the fundamentally different flow structures represented by the dual solutions. In the upper branch, the boundary layer profile remains stable and closely confined to the wall. When K increases, the porous medium becomes less permeable, enhancing the resistive drag exerted by the solid matrix on the fluid. This resistance suppresses cross-flow penetration and forces more fluid to adhere to the contracting sheet, thereby elevating the wall shear stress. Conversely, in the lower branch solution, the boundary layer is already weak and prone to separation; hence, increasing K weakens the near-wall momentum even further, resulting in reduced skin friction. Fig. 5 shows Nusselt number versus S for various K. For the upper branch solution, an increase is experienced in heat transfer rate when K is in-

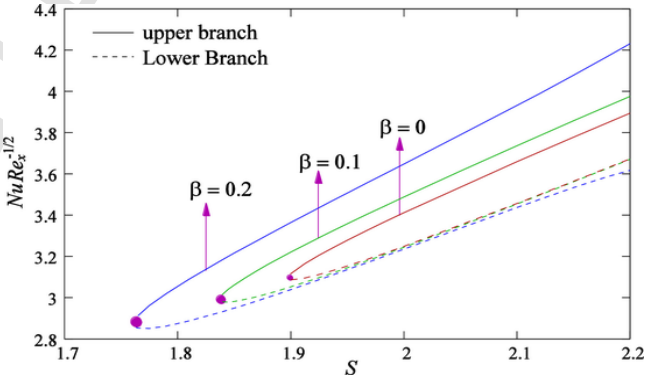


Fig. 3. Influence of Deborah number, β on $Nu\mathrm{Re}_x^{-1/2}$ with $K = 0.1$, $\mathrm{Pr} = 2$, $A^* = 0.1$, $B^* = 0.1$.

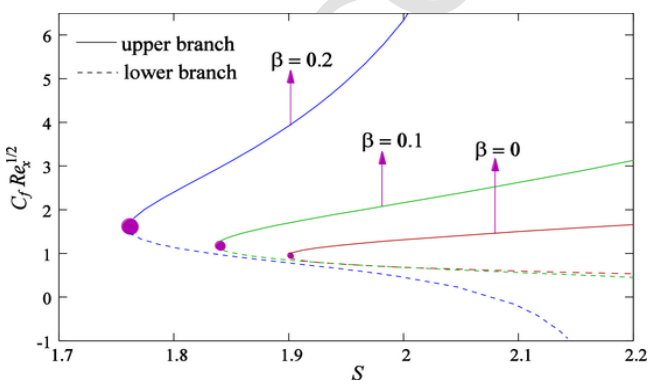


Fig. 2. Influence of Deborah number, β on $C_f\mathrm{Re}_x^{1/2}$ with $K = 0.1$, $\mathrm{Pr} = 2$, $A^* = 0.1$, $B^* = 0.1$.

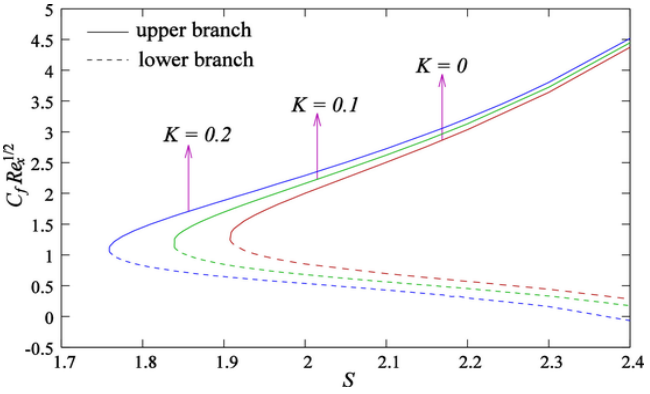


Fig. 4. Influence of permeability parameter, K on $C_f\mathrm{Re}_x^{1/2}$ with $\beta = 0.1$, $\mathrm{Pr} = 2$, $A^* = 0.1$, $B^* = 0.1$.

CORRECTED PROOF

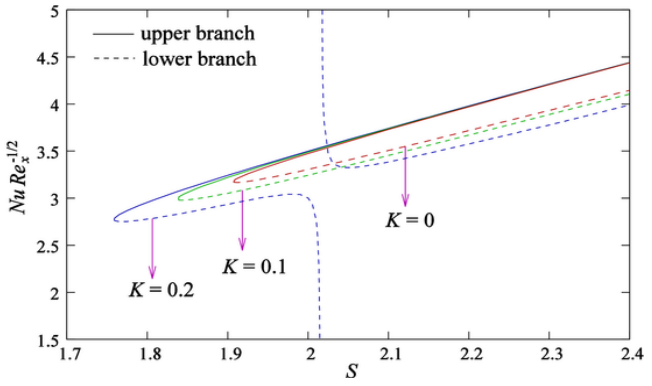


Fig. 5. Influence of permeability parameter, K on $Nu\mathrm{Re}_x^{-1/2}$ with $\beta = 0.1$, $\mathrm{Pr} = 2$, $A^* = 0.1$, $B^* = 0.1$.

creased, while the lower branch exhibits decreasing heat transfer. For the upper branch solution, the heat transfer rate rises with K, which is physically consistent: a more resistive porous medium limits fluid motion inside the matrix, reducing thermal diffusion away from the wall and strengthening the temperature gradient at the surface. In contrast, the lower branch solution shows a decreasing trend, due to less stable thermal boundary layer structure. Regarding $K = 0.2$, we note that there is a mass suction zone (between 2 and 2.1) where the cooling rate for the lower branch solution rises sharply, while near mass suction parameter value 2 the cooling rate for the lower branch solution experiences a rapid decline. Hence, porous media with lower permeability have a more pronounced influence on the heat transfer characteristics. For $K = 0$, i.e., in the absence of porous medium contracting sheet flow happens with the condition $S \geq 1.90751$ and the flow admits dual similarity solutions. When $K = 0.1$, flow field exhibits dual similarity solutions for $S \geq 1.8391$ and so no solution is in existence for $S < 1.8391$. For further increment, $K = 0.2$, dual similarity solutions exist with $S \geq 1.75895$ and no solution is found for $S < 1.75895$. It is clear from the aforementioned critical values of S that as K increases, the domain of existence of similarity solutions expands. This means that lower suction is sufficient to sustain the boundary layer flow, implying that as K increases, the amount of vorticity generated due to contraction of sheet within the boundary declines. Physically, increasing K implies enhanced resistance due to reduced permeability, which intensifies the interaction between the fluid and porous matrix, thus suppressing the vorticity generated by sheet contraction. Fig. 6 displays variation of the Nusselt number with S for various Pr. As anticipated, higher Pr causes growth in cooling rate in both solutions. Physically, higher Pr corresponds to lower thermal diffusivity. As a consequence, the thermal boundary layer contracts more rapidly than the velocity boundary layer. This contraction leads to a sharper temperature gradient at the wall. Because the Nusselt number scales directly with this gradient, an increase in Pr results in a markedly higher heat transfer rate. Fig. 7 and Fig. 8 depict the Nusselt number with S for the given parameter related to temperature dependent heat source/sink, namely A^*. When we increase $A^*(> 0$, heat source with $B^* > 0)$ and decrease the magnitude of $A^*(< 0$, heat sink with $B^* < 0)$, for upper branch, the heat transfer rate falls. Additionally, for similar variations in A^* for lower branch cooling rate exhibits identical behaviour; this displays a drop in heat transfer rate. Physically, this parameter characterizes internal volumetric heat generation or absorption that intensifies or weakens depending on the local fluid temperature. For $A^* > 0$, the medium functions as a temperature-dependent heat source, adding thermal energy as the fluid warms and thereby reducing wall cooling. For $A^* < 0$, it acts as a temperature-dependent heat sink; when the magnitude of A^* is reduced decreases, the sink weakens and removes less heat, which again lowers the boundary layer's cooling effectiveness. Fig. 9 and Fig. 10 show the Nusselt number fluctuations with S for various B^*. For both branches, the rate of heat transmission reduces when $B^*(> 0$, heat source with $A^* > 0)$ rises and decrease the magnitude of B^* $(< 0$, heat sink with $A^* < 0)$. For $B^* > 0$, a spatially varying heat source injects energy throughout the boundary layer, limiting the wall's ability of convective heat removal. This added energy en-

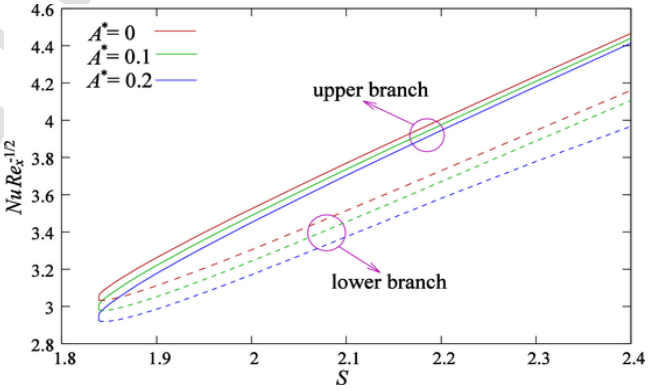


Fig. 7. Influence of temperature-dependent heat source, $A^*(>0)$ on $Nu\mathrm{Re}_x^{-1/2}$ with $\beta = 0.1$, $K = 0.1$, $\mathrm{Pr} = 2$, $B^* = 0.1$.

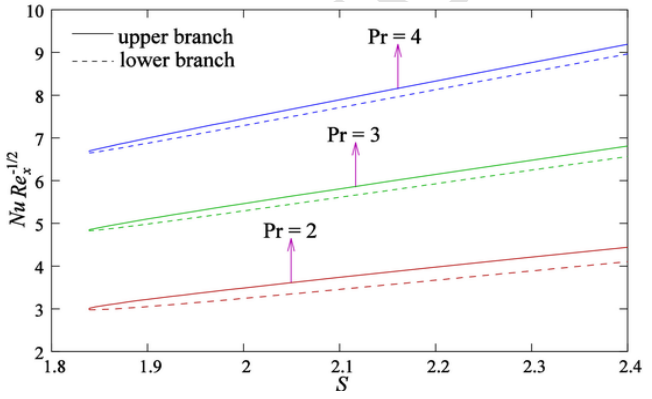


Fig. 6. Influence of Prandtl number, Pr on $Nu\mathrm{Re}_x^{-1/2}$ with $\beta = 0.1$, $K = 0.1$, $A^* = 0.1$, $B^* = 0.1$.

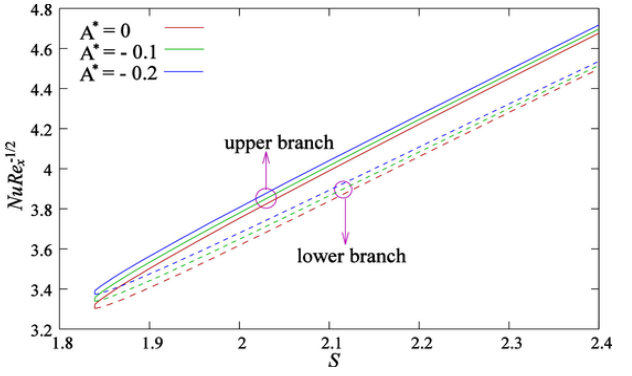


Fig. 8. Influence of temperature-dependent heat sink, $A^*(<0)$ on $Nu\mathrm{Re}_x^{-1/2}$ with $\beta = 0.1$, $K = 0.1$, $\mathrm{Pr} = 2$, $B^* = -0.1$.

CORRECTED PROOF

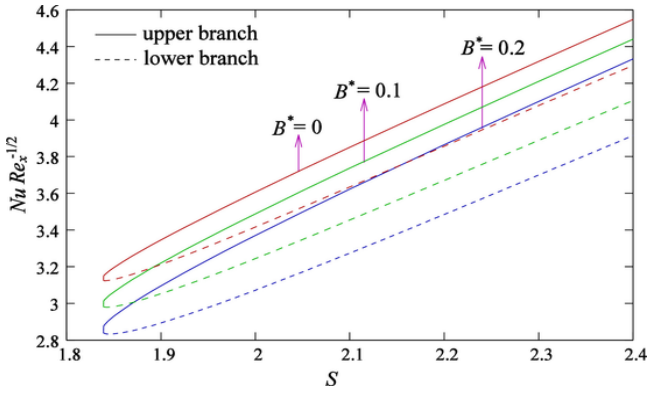


Fig. 9. Influence of space-dependent heat source, B^* (>0) on $Nu\mathrm{Re}_x^{-1/2}$ with $\beta = 0.1$, $K = 0.1$, Pr $= 2$, $A^* = 0.1$.

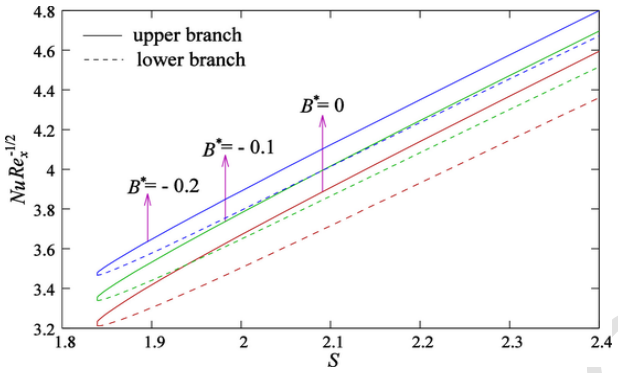


Fig. 10. Influence of space-dependent heat sink, $B^*(<0)$ on $Nu\mathrm{Re}_x^{-1/2}$ with $\beta = 0.1$, $K = 0.1$, Pr $= 2$, $A^* = -0.1$.

larges the thermal layer, flattens the wall temperature gradient, and thus reduces the Nusselt number for both solution branches. For $B^* < 0$, a spatially dependent heat sink removes energy as the distance from the wall increases; when the magnitude of B^* becomes smaller, its cooling action weakens, the fluid near the wall becomes warmer, the temperature gradient drops, and the Nusselt number decreases once more. All these observations are further supported by the numerical results listed in Table 1 and Table 2.

4.2. Impacts on velocity and temperature

Fig. 11 depicts the variations in velocity distributions for various Deborah number β. The two solution branches remain clearly distinguishable. In every case, the velocity boundary layer associated with the lower branch is noticeably thicker than that of the upper branch. For the upper branch solution, an increase in β enhances the fluid velocity at any fixed location, thereby thinning the boundary layer. In contrast, for the lower branch, velocity initially decreases with β, but beyond a certain threshold the velocity begins to rise, leading to a corresponding reduction in boundary layer thickness(BLT). This behaviour is physically consistent with the rheology of Maxwell fluids: higher Deborah numbers reflect stronger elastic effects and reduced effective fluidity, which suppress the formation of thicker boundary layers. Actually, a larger Deborah number implies that the fluid's relaxation time becomes comparable to the characteristic deformation time, enabling the elastic stresses to store and release energy more effectively and this stored elastic energy assists the flow near the contracting surface, accelerates the fluid, and thereby compresses the boundary layer. Fig. 12 highlights the thermal response to changes in β. For the upper branch, an increase in β results in a reduction of the thermal boundary layer thickness, indicating more efficient heat removal. In the lower branch, however, the thermal boundary layer becomes thicker as β rises. The reduced fluidity at high Deborah numbers enhances heat-transfer capability for the upper branch, while the weaker flow intensity in the lower branch permits thermal diffusion to dominate, causing the thermal layer to expand. Fig. 13 demonstrates that the velocity $f'(\eta)$ rises with permeability parameter K for upper branch, accompanied by a thinning of the momentum BLT, and for the lower branch $f'(\eta)$ decreases with K, momentum BLT being thicker. The temperature distributions in Fig. 14 display contrasting trends: in the upper branch, both temperature and thermal boundary layer thickness decrease with increasing K, whereas the opposite occurs for the lower branch. This is attributed to the interaction between fluid momentum and the porous matrix: reduced permeability enhances resistance to flow, modifies convective strength, and therefore influences heat diffusion. As expected, the lower branch

Table 1
Variations in $C_f\mathrm{Re}_x^{1/2}$ for various values S, K and β.

S	K	β	$C_f\mathrm{Re}_x^{1/2}$	
			Upper branch solution	Lower Branch solution
1.85	0.1	0.1	1.38936949	1.02155828
1.90	0.1	0.1	1.69646847	0.84821391
1.95	0.1	0.1	1.93523180	0.75244081
2.0	0.1	0.1	2.16110373	0.67993534
2.05	0.1	0.1	2.38746357	0.61844420
2.10	0.1	0.1	2.62119412	0.56244284
2.15	0.1	0.1	2.86722875	0.50851816
2.2	0.1	0.1	3.12997127	0.45402074
2.3	0.1	0.1	3.72404718	0.33319199
2.4	0.1	0.1	4.44764614	0.17611793
1.95	0.1	0	1.20001829	0.74850744
1.95	0.1	0.15	2.77435660	0.71841216
1.95	0.1	0.17	3.36044216	0.68611372
1.95	0.1	0.2	4.93494320	0.58865333
1.95	0	0.1	1.72368217	0.97465098
1.95	0.13	0.1	1.98493433	0.69930309
1.95	0.16	0.1	2.03093195	0.64972967
1.95	0.2	0.1	2.08774567	0.58799291

Table 2
Variations in $Nu\mathrm{Re}_x^{-1/2}$ for various values S, K, β, A^*, B^* and Pr.

S	K	β	Pr	A^*	B^*	$Nu\mathrm{Re}_x^{-1/2}$	
						Upper branch solution	Lower Branch solution
1.85	0.1	0.1	2	0.1	0.1	−3.06172347	−2.98194075
1.85	0.1	0.1	3	0.1	0.1	−4.90290594	−4.84710741
1.85	0.1	0.1	4	0.1	0.1	−6.75169420	−6.70838690
1.85	0.1	0.1	6.2	0.1	0.1	−10.82731247	−10.79692745
1.87	0.1	0.1	2	0.1	0.1	−3.13026404	−3.00498819
1.90	0.1	0.1	2	0.1	0.1	−3.22041154	−3.05239868
1.95	0.1	0.1	2	0.1	0.1	−3.35807824	−3.14427924
2.0	0.1	0.1	2	0.1	0.1	−3.48811340	−3.24386621
2.05	0.1	0.1	2	0.1	0.1	−3.61371231	−3.34764528
2.1	0.1	0.1	2	0.1	0.1	−3.73633313	−3.45394421
2.15	0.1	0.1	2	0.1	0.1	−3.85678411	−3.56175017
2.2	0.1	0.1	2	0.1	0.1	−3.97558188	−3.67040396
1.95	0.1	0.1	2	0	0.1	−3.39669991	−3.20475292
1.95	0.1	0.2	2	0.2	0.1	−3.31808209	−3.07377934
1.95	0.1	0.1	2	0.1	0	−3.47900081	−3.31088138
1.95	0.1	0.1	2	0.1	0.2	−3.23715425	−2.97767615
1.95	0.1	0.1	2	0	−0.1	−3.62944603	−3.50963855
1.95	0.1	0.1	2	−0.1	−0.1	−3.65858912	−3.54229331
1.95	0.1	0.1	2	−0.2	−0.1	−3.68733811	−3.57484436
1.95	0.1	0.1	2	−0.1	0	−3.54633808	−3.40064454
1.95	0.1	0.1	2	−0.1	−0.2	−3.77083302	−3.68394065

UNCORRECTED PROOF

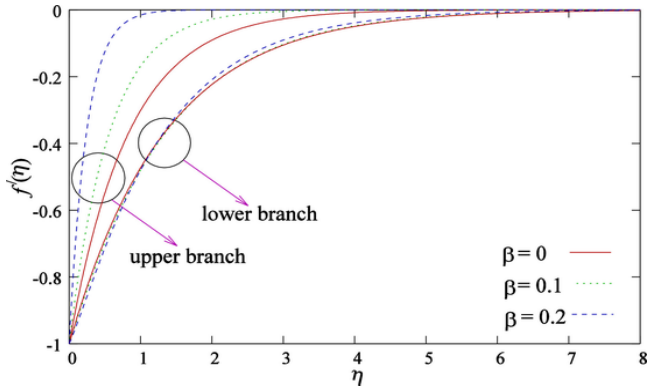


Fig. 11. Effect of Deborah number, β on $f'(\eta)$ with $K = 0.1$, $S = 1.95$, Pr = 2, $A^* = 0.1$, $B^* = 0.1$.

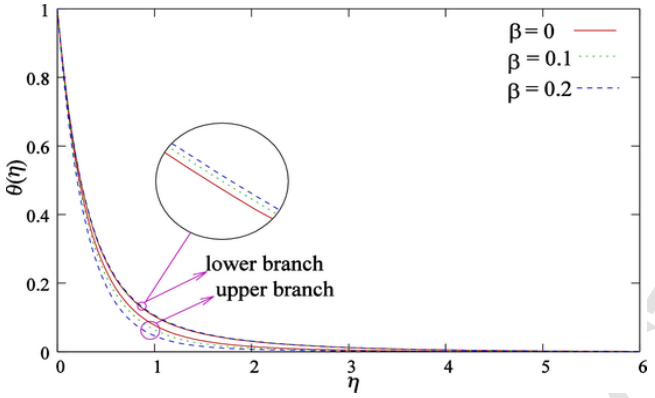


Fig. 12. Effect of Deborah number, β on $\theta(\eta)$ with $K = 0.1$, $S = 1.95$, Pr = 2, $A^* = 0.1$, $B^* = 0.1$.

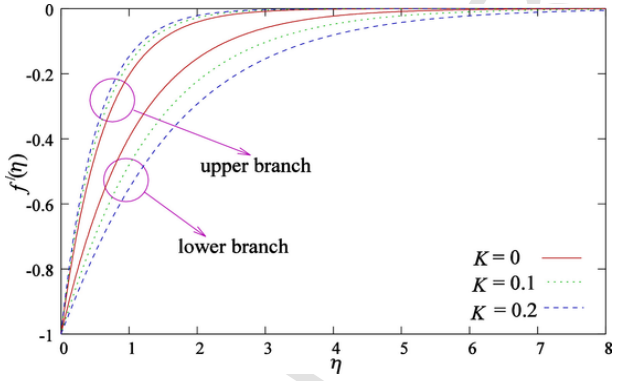


Fig. 13. Effect of permeability parameter, K on $f'(\eta)$ with $\beta = 0.1$, $S = 1.95$, Pr = 2, $A^* = 0.1$, $B^* = 0.1$.

profiles consistently exhibit thicker boundary layers than the upper branch profiles. Overall, the porous medium plays a dual role: it modulates the effective resistance to fluid motion and indirectly governs the balance between convection and diffusion. The upper branch solution, being convection-dominated, responds by accelerating and cooling as permeability parameter increases, while the lower branch solution, being diffusion-dominated, exhibits thicker boundary layers and weaker cooling performance. These contrasting behaviors underscore the physical significance of dual solutions in viscoelastic porous flows.

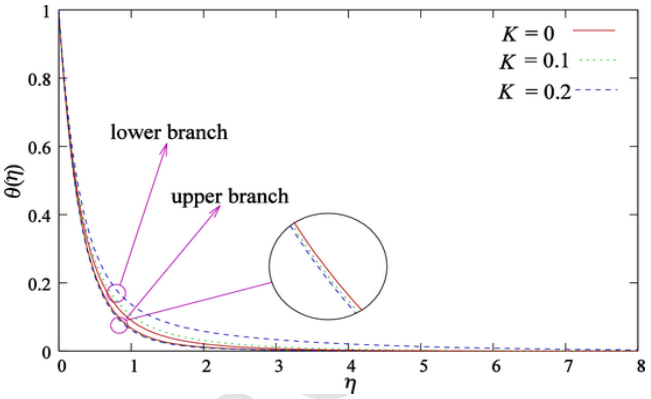


Fig. 14. Effect of permeability parameter, K on $\theta(\eta)$ with $\beta = 0.1$, $S = 1.95$, Pr = 2, $A^* = 0.1$, $B^* = 0.1$.

Fig. 15 reveals the variation of temperature with Prandtl number Pr. In both solution branches, higher values of Pr lead to lower temperatures and thinner thermal boundary layers. This is physically intuitive: increasing Pr corresponds to a reduction in thermal diffusivity, which restricts the spread of heat away from the wall and compresses the thermal layer. Mass suction is a crucial flow-control mechanism, particularly because it mitigates vorticity generated by surface contraction and helps sustain the boundary layer. Fig. 16 displays the velocity profiles for various suction parameter values ($S > 0$). In the upper branch, increasing suction strengthens the near-wall velocity and reduces boundary layer thickness. Conversely, in the lower branch, velocity diminishes with increasing suction, and the boundary layer becomes thicker. Temperature profiles in Fig. 17 show that thermal BLT falls with suction for the upper branch, but it rises slightly for the lower branch. These trends follow classical boundary layer theory: suction draws the fluid layers near to the surface, compressing the thickness of the boundary layer, whereas injection (negative suction) has the opposite effect: it pushes the fluid layers outward thickening boundary layer. Overall, the physical trends reflect classical boundary layer behavior but are modified by the presence of dual solutions. In the upper branch, suction energizes the flow, sharpens gradients, and strengthens both velocity and thermal field confinement—leading to thinner boundary layers and more efficient cooling. In the lower branch, suction reduces overall flow strength, allowing diffusion to dominate and resulting in thicker boundary layers and poorer transport performance. These contrasting responses highlight the sensitivity of viscoelastic boundary layer flows

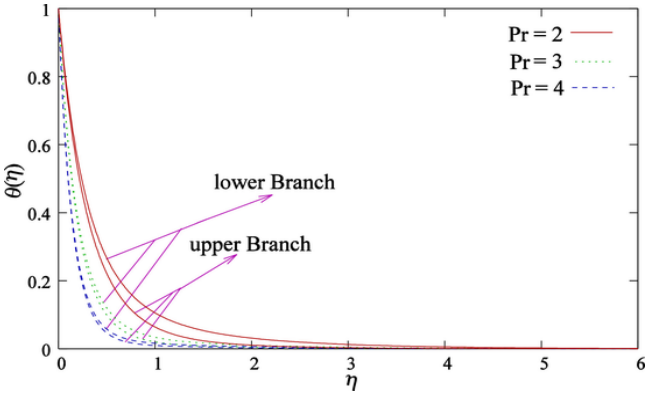


Fig. 15. Effect of Prandtl number, Pr on $\theta(\eta)$ with $\beta = 0.1$, $K = 0.1$, $S = 1.95$, $A^* = 0.1$, $B^* = 0.1$.

UNCORRECTED PROOF

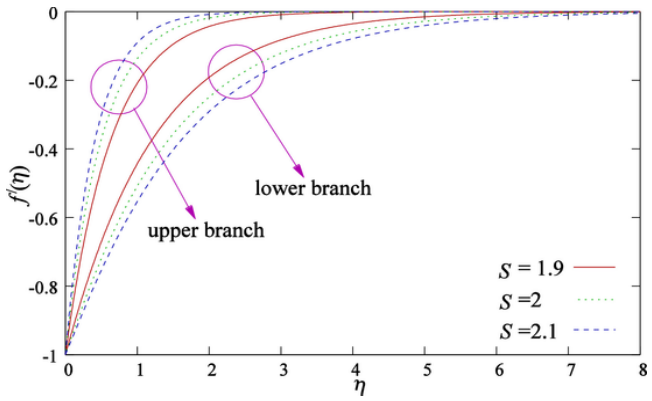


Fig. 16. Effect of mass suction parameter, S on $f'(\eta)$ with $\beta = 0.1$, $K = 0.1$, $\mathrm{Pr} = 2$, $A^* = 0.1$, $B^* = 0.1$.

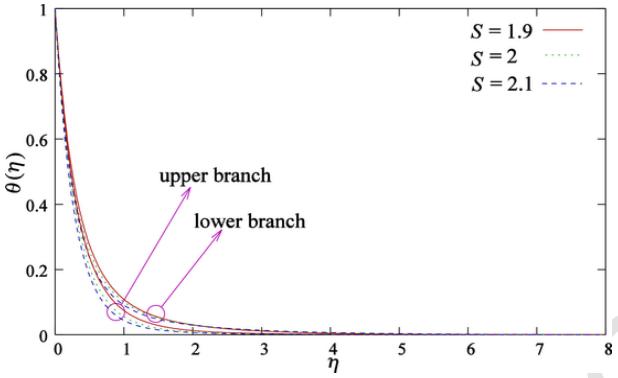


Fig. 17. Effect of mass suction parameter, S on $\theta(\eta)$ with $\beta = 0.1$, $K = 0.1$, $\mathrm{Pr} = 2$, $A^* = 0.1$, $B^* = 0.1$.

to mass transfer and the crucial role suction plays in stabilizing or destabilizing different solution branches. Fig. 18 and Fig. 19 deliver influence of the temperature-dependent heat source/sink parameter A^* on temperature. It shows that dimensionless temperature $\theta(\eta)$ upsurges with positive A^*(temperature-dependent heat source) and drops with negative A^*(heat sink being temperature dependent) for both solution branches. This indicates that positive A^* diminishes the cooling capacity at the wall, while negative A^* enhances it. Moreover, the upper

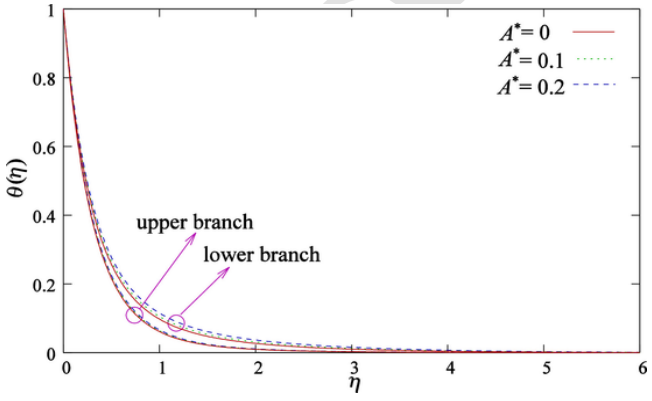


Fig. 18. Effect of temperature-dependent heat source, $A^*(>0)$ on $\theta(\eta)$ with $\beta = 0.1$, $K = 0.1$, $S = 1.95$, $\mathrm{Pr} = 2$, $B^* = 0.1$.

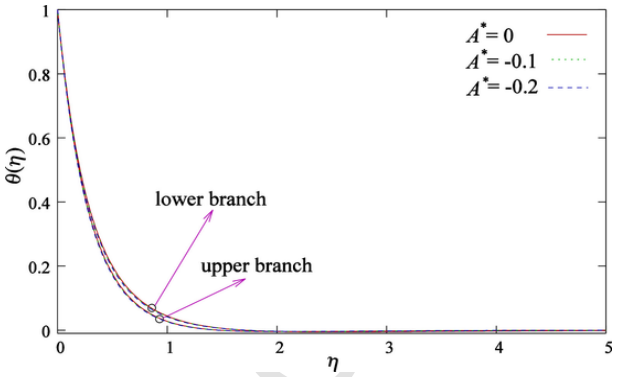


Fig. 19. Effect of temperature-dependent heat sink, $A^*(<0)$ on $\theta(\eta)$ with $\beta = 0.1$, $K = 0.1$, $S = 1.95$, $\mathrm{Pr} = 2$, $B^* = -0.1$.

branch solution is most sensitive to A^* near the contracting surface, whereas in the lower branch the strongest thermal response occurs farther from the wall within the boundary layer. The two branches respond differently due to their distinct flow structures: the upper branch is convection-dominated, so its temperature is most sensitive to A^* near the contracting surface. The lower branch is more diffusion-dominated, causing the strongest influence of A^* to appear farther from the wall within the thermal boundary layer. The effects of the space-dependent heat source/sink parameter B^* are presented in Fig. 20 and Fig. 21.

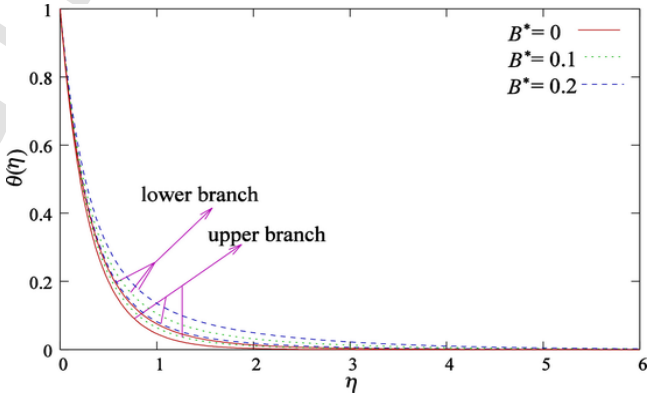


Fig. 20. Effect of space-dependent heat source, $B^*(>0)$ on $\theta(\eta)$ with $\beta = 0.1$, $K = 0.1$, $S = 1.95$, $\mathrm{Pr} = 2$, $A^* = 0.1$.

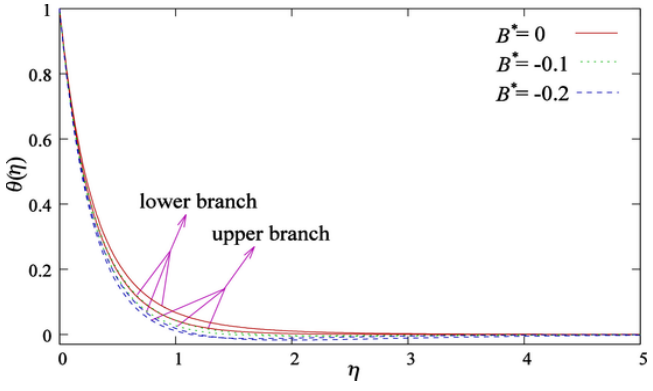


Fig. 21. Effect of space-dependent heat sink, $B^*(<0)$ on $\theta(\eta)$ with $\beta = 0.1$, $K = 0.1$, $S = 1.95$, $\mathrm{Pr} = 2$, $A^* = -0.1$.

Temperature enhances as positive B^* grows (heat source) and decreases with larger magnitudes of negative B^* (heat sink) for both solution branches. Furthermore, as $B^* > 0$ (heat source) increases, the thermal BLT increases, with higher temperature inside boundary layer. Physically, a space-dependent heat source injects energy into the fluid throughout the boundary layer, elevating the temperature, whereas a sink extracts heat and lowers the temperature. Notably, the influence of B^* on thermal behaviour is more pronounced than that of A^*, indicating that spatially distributed heat addition or removal exerts stronger control over thermal transport than temperature-dependent mechanisms.

5. Flow stability

The numerical results of the BVP (13)-(14) subject to conditions (15) confirm the existence of dual solutions under appropriate flow situations. In order to determine physical nature of non-unique solutions, following the methodology of Weidman et al. [56] and Harris et al. [57] a stability analysis is conducted and this analysis assesses whether each solution branch is stable or unstable to small temporal disturbances. For this purpose of stability analysis, a dimensionless time-dependent variable $\tau = \frac{u_w}{x}t$ is introduced. For unsteady flow, the governing momentum and energy equations can be reformulated as [58,59]:

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + \lambda\left[u^2\frac{\partial^2 u}{\partial x^2} + v^2\frac{\partial^2 u}{\partial y^2} + 2uv\frac{\partial^2 u}{\partial x \partial y}\right] = \upsilon\frac{\partial^2 u}{\partial y^2} - \frac{\upsilon}{k}u, \tag{19}$$

$$\frac{\partial T}{\partial t} + u\frac{\partial T}{\partial x} + v\frac{\partial T}{\partial y} = \frac{\kappa}{(\rho c_p)}\frac{\partial^2 T}{\partial y^2} + \frac{\kappa}{(\rho c_p)}\frac{U_w}{\upsilon x}\left[A^*\left(T - T_\infty\right) + B^*\left(T_w - T_\infty\right)e^{-\frac{y}{x}\sqrt{\frac{U_w x}{\upsilon}}}\right], \tag{20}$$

with the boundary restrictions

$$u = U_w = -cx,\ v = -v_w,\ T = T_w \text{ at } y = 0, \tag{21}$$

$$u \to 0\ T = T_\infty \text{ as } y \to \infty. \tag{22}$$

To derive the dimensionless unsteady formulation, the following time-dependent transformations are introduced:

$$\psi = \sqrt{x\upsilon U_w}F(\eta,\tau), \Theta(\eta,\tau) = \frac{T - T_\infty}{T_w - T_\infty},\ \eta = \sqrt{\frac{U_w}{\upsilon x}}y,\ \tau = \frac{U_w}{x}t. \tag{23}$$

Substituting (23) into Eqs. (19)-(22) yields the transformed equations along with corresponding boundary conditions as:

$$F_{\eta\eta\eta} + \beta\left(2FF_\eta F_{\eta\eta} - F^2F_{\eta\eta\eta}\right) + FF_{\eta\eta} - {F_\eta}^2 - KF_\eta - F_{\eta\tau} = 0, \tag{24}$$

$$\Theta_{\eta\eta} + \mathrm{Pr}F\Theta_\eta + A^*\Theta + B^*e^{-\eta} - \mathrm{Pr}\Theta_\tau = 0, \tag{25}$$

with

$$\left.\begin{aligned} &F(0,\tau) = S, F_\eta(0,\tau) = -1,\ F_\eta(\infty,\tau) = 0,\\ &\Theta(0,\tau) = 1 \text{ and } \Theta(\infty,\tau) = 0. \end{aligned}\right\} \tag{26}$$

Now, to check the behaviour regarding the stability, the solutions of above BVP, (13)-(15) are perturbed by small disturbances as:

$$\left.\begin{aligned} &F(\eta,\tau) = f(\eta) + e^{-\alpha\tau}g(\eta,\tau),\\ &\Theta(\eta,\tau) = \theta(\eta) + e^{-\alpha\tau}\phi(\eta,\tau). \end{aligned}\right\} \tag{27}$$

Here, $g(\eta,\tau)$ and $\phi(\eta,\tau)$ denote the small perturbations introduced in the solution of the steady-state problem and α is the disturbance growth (or decay) rate. For linear stability analysis of the problem, we linearized the equations with respect to these perturbations and we get:

$$g_{\eta\eta\eta} + \beta\left\{2\left(ff_\eta g_{\eta\eta} + f_\eta f_{\eta\eta}g + ff_{\eta\eta}g_\eta\right) - f^2g_{\eta\eta\eta} - 2ff_{\eta\eta\eta}g\right\} + f_{\eta\eta}g + fg_{\eta\eta} - 2f_\eta g_\eta - Kg_\eta - g_{\eta\tau} + \alpha g_\eta = 0, \tag{28}$$

$$\phi_{\eta\eta} + \mathrm{Pr}\left(f\phi_\eta + \theta_\eta g\right) + A^*\phi - \mathrm{Pr}\phi_\tau + \mathrm{Pr}\alpha\phi = 0, \tag{29}$$

with corresponding BCs.

$$\left.\begin{aligned} &g(0,\tau) = 0,\ g_\eta(0,\tau) = 0,\ \phi(0,\tau) = 0,\\ &g_\eta(\infty,\tau) = 0,\ \phi(\infty,\tau) = 0. \end{aligned}\right\} \tag{30}$$

Following the approach of Weidman et al. [56], the stability of the steady solutions is examined by setting $\tau = 0$ in Eqs. (28)-(30). So, quantities $g(\eta,0) = g_0(\eta)$ and $\phi(\eta,0) = \phi_0(\eta)$ represent the initial disturbance to the solution. Hence, a reduced system can be expressed as:

$$g''_0 + \beta\left\{2\left(ff'g''_0 + ff''g_0 + ff''g'_0\right) - f^2g''_0 - 2ff''g_0\right\} + f''g_0 + fg''_0 - 2f'g'_0 - Kg'_0 + \alpha g'_0 = 0, \tag{31}$$

$$\phi''_0 + \mathrm{Pr}\left(f\phi'_0 + \theta'g_0\right) + A^*\phi_0 + \mathrm{Pr}\alpha\phi_0 = 0, \tag{32}$$

subjected to boundary conditions

$$\left.\begin{aligned} &g_0(0) = 0,\ g'_0(0) = 0,\ \phi_0(0) = 0,\\ &g'_0(\infty) = 0,\ \phi_0(\infty) = 0. \end{aligned}\right\} \tag{33}$$

It is important to observe that the resulting formulation constitutes an eigenvalue problem, yielding an infinite set of eigenvalues, $\alpha_i,\ i = 1, 2, \ldots$ satisfying $\alpha_1 < \alpha_2 < \alpha_3 < \ldots$. The stability of computed steady solution is decided by sign of the smallest (leading) eigenvalue. A positive leading eigenvalue corresponds to stable solution, because it indicates a situation where disturbances decay with time, while the negative sign eigenvalue confirms that the corresponding steady solution is unstable due to the growth in introduced disturbances.

Solving Eqs. (31) and (32) subject to the boundary conditions (33), the smallest eigenvalues are obtained for both solutions, as reported in Table 3. Exhibited positive eigenvalues for upper solution branches indicate the physical stability of the solutions and oppositely, negative eigenvalues for lower solution branches specify the physical instability of the solutions. It should be emphasized, however, that linear stability alone does not guarantee that a solution will manifest in a real experimental setting. So, values of both solutions in this investigation are identical in physical sense and stable and unstable branches will help to comment on the outcome of real experiment.

6. Concluding notes

The present work explores the heat transfer behaviour of a Maxwell fluid subjected to a porous contracting surface within a porous medium, incorporating spatially and temperature-dependent heat source/sink effects. The major conclusions drawn from this investigation are outlined below:

- Maxwell fluid elasticity reduces vorticity production, allowing the boundary layer to remain attached with significantly less suction compared to Newtonian fluids. So, higher Deborah numbers expand the domain of similarity solutions, because elastic stresses suppress boundary layer separation and stabilize the flow.

Table 3
Smallest eigenvalue, α_1 for some S with $\beta = 0.1$, $K = 0.1$, $A^* = 0.1$, $\delta_2 = 0$ and $\mathrm{Pr} = 2$.

S	α_1	
	Upper branch solution	Lower branch solution
2	0.8810518	−0.0376726
2.1	1.1498579	−0.1290175
2.2	1.4609727	−0.2006274
2.3	1.8396721	−0.2601093
2.4	2.3116364	−0.3115262

UNCORRECTED PROOF

- Skin friction increases with Deborah number for the upper branch but decreases for the lower branch, reflecting stronger near-wall momentum in the stable branch and weaker momentum in the unstable branch.
- Nusselt number increases with Deborah number in the upper branch, demonstrating superior cooling capability of Maxwell fluids relative to Newtonian fluids.
- Increasing permeability parameter K enlarges the existence range of similarity solutions, indicating that reduced permeability suppresses vorticity from sheet contraction.
- For the upper branch, skin friction and heat transfer rise with increasing K, as porous resistance enhances near-wall shear and steepens the thermal gradient. Whereas, for the lower branch, skin friction and Nusselt number decrease with higher K, due to weakened near-wall momentum and a more diffusion-dominated thermal structure.
- Higher Prandtl numbers yield stronger cooling because low thermal diffusivity compresses the thermal boundary layer and increases wall temperature gradients.
- Temperature-dependent heat sources consistently reduce heat transfer, while temperature-dependent sinks enhance cooling for both branches.
- Spatially dependent heat sources raise fluid temperature throughout the boundary layer, producing thicker thermal layers and reduced heat transfer rates. Whereas, spatial heat sinks strengthen wall cooling, as heat is continuously extracted through the boundary layer thickness.
- Increasing the permeability parameter K enhances the fluid velocity in the upper branch solution but reduces it in the lower branch; conversely, temperature decreases with K for the upper branch and increases for the lower branch.
- The thermal field is more strongly influenced by space-dependent heat source/sink effects than by temperature-dependent heat generation or absorption.
- Suction ($S > 0$) stabilizes and sharpens the boundary layer in the upper branch, strengthening velocity near the wall and reducing both momentum and thermal BLT.
- For the lower branch, suction weakens velocity and enlarges boundary layers, reflecting its sensitivity to mass transfer and its inherent instability compared to the upper branch.
- The upper-branch solutions are physically stable, as indicated by their positive smallest eigenvalues, whereas the lower-branch solutions are unstable, corresponding to negative eigenvalues. Linear stability does not guarantee physical realizability, meaning that despite the mathematical stability or instability, both solution branches must still be interpreted cautiously when predicting real experimental outcomes, with each branch offering insight into possible physical behaviours.

CRediT authorship contribution statement

Amit Kumar Pandey: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Prakash Goswami:** Writing – review & editing, Visualization, Validation, Methodology, Investigation, Formal analysis. **Sohita Rajput:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Krishnendu Bhattacharyya:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Methodology, Investigation, Formal analysis, Conceptualization. **Ali J. Chamkha:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis. **Dhananjay Yadav:** Writing – review & editing, Visualization, Validation, Methodology, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors sincerely appreciate the valuable comments and suggestions provided by the reviewers. K. Bhattacharyya acknowledges "Faculty Incentive Grant" (No. R/Dev/D/IoE/Incentive/2022-23/47686) of Banaras Hindu University.

Data availability

No data was used for the research described in the article.

References

[1] A. Bejan, Convection Heat Transfer, John wiley & sons, 2013.
[2] A. Acrivos, On the combined effect of forced and free convection heat transfer in laminar boundary layer flows, Chem. Eng. Sci. 21 (4) (1966) 343–352.
[3] M. Fathizadeh, F. Rashidi, Boundary layer convective heat transfer with pressure gradient using Homotopy Perturbation Method (HPM) over a flat plate, Chaos Solit. Fractals 42 (4) (2009) 2413–2419.
[4] N.C. Roşca, A.V. Roşca, Pop I. Stagnation point flow and heat transfer over a non-linearly moving flat plate in a parallel free stream with slip, Commun. Nonlinear Sci. Numer. Simul. 19 (6) (2014) 1822–1835.
[5] S.K. Parida, S. Panda, B.R. Rout, MHD boundary layer slip flow and radiative nonlinear heat transfer over a flat plate with variable fluid properties and thermophoresis, Alex. Eng. J. 54 (4) (2015) 941–953.
[6] M. Sohail, U. Nazir, Y.M. Chu, H. Alrabaiah, W. Al-Kouz, P. Thounthong, Computational exploration for radiative flow of Sutterby nanofluid with variable temperature-dependent thermal conductivity and diffusion coefficient, Open Phys. 18 (1) (2020) 1073–1083.
[7] T.A. Yusuf, F. Mabood, B.C. Prasannakumara, Sarris IE. Magneto-bioconvection flow of Williamson nanofluid over an inclined plate with gyrotactic microorganisms and entropy generation, Fluids 6 (3) (2021) 109.
[8] D. Qaiser, Z. Zheng, M.R. Khan, Numerical assessment of mixed convection flow of Walters-B nanofluid over a stretching surface with Newtonian heating and mass transfer, Therm. Sci. Eng. Prog. 22 (2021) 100801.
[9] T.A.M. Shatnawi, N. Abbas, W. Shatanawi, Mathematical analysis of unsteady stagnation point flow of radiative Casson hybrid nanofluid flow over a vertical Riga sheet, Mathematics 10 (19) (2022) 3573.
[10] J.C. Maxwell, On the dynamical theory of gases, Philos. Trans. R. Soc. 157 (1867) 49–88.
[11] D. Vieru, W. Akhtar, C. Fetecau, C. Fetecau, Starting solutions for the oscillating motion of a Maxwell fluid in cylindrical domains, Meccanica 42 (6) (2007) 573–583.
[12] T. Hayat, C. Fetecau, Z. Abbas, N. Ali, Flow of a Maxwell fluid between two side walls due to a suddenly moved plate, Nonlinear Anal.: Real World Appl. 9 (5) (2008) 2288–2295.
[13] F. Olsson, J. Yström, Some properties of the upper convected Maxwell model for viscoelastic fluid flow, J. Nonnewton. Fluid. Mech. 48 (1–2) (1993) 125–145.
[14] J.J. Choi, Z. Rusak, J.A. Tichy, Maxwell fluid suction flow in a channel, J. Nonnewton. Fluid. Mech. 85 (2–3) (1999) 165–187.
[15] C. Fetecau, C. Fetecau, A new exact solution for the flow of a Maxwell fluid past an infinite plate, Int. J. Non. Linear. Mech. 38 (3) (2003) 423–427.
[16] C. Fetecau, C. Fetecau, The Rayleigh–Stokes-Problem for a fluid of Maxwellian type, Int. J. Nonlinear Mech. 38 (4) (2003) 603–607.
[17] S. Mukhopadhyay, K. Bhattacharyya, Unsteady flow of a Maxwell fluid over a stretching surface in presence of chemical reaction, J. Egypt. Math. Soc. 20 (3) (2012) 229–234.
[18] X.H. Zhang, R. Shah, S. Saleem, N.A. Shah, Z.A. Khan, J.D. Chung, Natural convection flow Maxwell fluids with generalized thermal transport and newtonian heating, Case Stud. Therm. Eng. 27 (2021) 101226.
[19] F.T. Akyildiz, F. Alshammari, A new analysis of Galerkin Legendre spectral methods for coupled hyperbolic/parabolic system arising in unsteady MHD flow of Maxwell fluid and numerical simulation, Appl. Numer. Math. 176 (2022) 83–103.
[20] M. Miklavčič, C. Wang, Viscous flow due to a shrinking sheet, Q. Appl. Math. 64 (2) (2006) 283–290.
[21] M. Sajid, T. Hayat, The application of homotopy analysis method for MHD viscous flow due to a shrinking sheet, Chaos Solit. Fractals 39 (3) (2009) 1317–1323.
[22] T. Hayat, Z. Abbas, M. Sajid, On the analytic solution of magnetohydrodynamic flow of a second grade fluid over a shrinking sheet, J. Appl. Mech. 74 (6) (2007) 1165–1171.
[23] M. Sajid, T. Javed, T. Hayat, MHD rotating flow of a viscous fluid over a shrinking surface, Nonlinear. Dyn. 51 (1) (2008) 259–265.
[24] A. Ishak, Y.Y. Lok, Pop I. Non-Newtonian power-law fluid flow past a shrinking

UNCORRECTED PROOF

sheet with suction, Chem. Eng. Commun. 199 (1) (2012) 142–150.
[25] M. Turkyilmazoglu, Multiple solutions of hydromagnetic permeable flow and heat for viscoelastic fluid, J. Thermophys. Heat. Trans. 25 (4) (2011) 595–605.
[26] C.Y. Wang, Stagnation flow towards a shrinking sheet, Int. J. Nonlinear Mech. 43 (2008) 377–382.
[27] A. Ishak, Y.Y. Lok, Pop I. Stagnation-point flow over a shrinking sheet in a micropolar fluid, Chem. Eng. Commun. 197 (2010) 1417–1427.
[28] K. Bhattacharyya, S. Mukhopadhyay, G.C. Layek, Slip effects on boundary layer stagnation-point flow and heat transfer towards a shrinking sheet, Int. J. Heat. Mass Transf. 54 (2011) 308–313.
[29] K. Bhattacharyya, G.C. Layek, Effects of suction/blowing on steady boundary layer stagnation-point flow and heat transfer towards a shrinking sheet with thermal radiation, Int. J. Heat. Mass Transf. 54 (2011) 302–307.
[30] N.S. Ismail, N.M. Arifin, R. Nazar, N. Bachok, Stability analysis of unsteady MHD stagnation point flow and heat transfer over a shrinking sheet in the presence of viscous dissipation, Chin. J. Phys. 57 (2019) 116–126.
[31] K. Bhattacharyya, M.G. Arif, W. Ali Pk, MHD boundary layer stagnation-point flow and mass transfer over a permeable shrinking sheet with suction/blowing and chemical reaction, Acta Tech. CSAV 57 (1) (2012) 1–15.
[32] S. Mukhopadhyay, K. Bhattacharyya, G.C. Layek, Mass transfer over an exponentially stretching porous sheet embedded in a stratified medium, Chem. Eng. Commun. 201 (2) (2014) 272–286.
[33] L.T. Benos, K.R. Nagaraju, U.S. Mahabaleshwar, M.S. Prasad, I.E. Sarris, G. Lorenzini, Magnetohydrodynamic and radiation effects on the heat transfer of a continuously stretching/shrinking sheet with mass transpiration of the horizontal boundary, Chin. J. Phys. 72 (2021) 700–715.
[34] S. Rajput, A.K. Pandey, K. Bhattacharyya, I. Pop, Effect of convective boundary condition on unsteady flow of CNT-H_2O nanofluid towards a stagnation-point on a shrinking/expanding flat sheet, Proc. Inst. Mech. Eng. E: J. Process Mech. Eng. 236 (3) (2022) 1023–1033.
[35] S. Nadeem, B. Ishtiaq, N. Abbas, Impact of thermal radiation on two-dimensional unsteady third-grade fluid flow over a permeable stretching Riga plate, Int. J. Mod. Phys. B 37 (1) (2023) 2350009.
[36] N. Abbas, W. Shatanawi, T.A.M. Shatnawi, Transportation of nanomaterial Maxwell fluid flow with thermal slip under the effect of Soret–Dufour and second–order slips: nonlinear stretching, Sci. Rep. 13 (2023) 2182.
[37] S. Nadeem, A. Amin, N. Abbas, On the stagnation point flow of nanomaterial with base viscoelastic micropolar fluid over a stretching surface, Alex. Eng. J. 59 (3) (2020) 1751–1760.
[38] N. Abbas, M. Ali, W. Shatanawi, Chemical reactive second-grade nanofluid flow past an exponential curved stretching surface: Numerically, Int. J. Mod. Phys. B 38 (2) (2025) 2450026.
[39] S. Rajput, K. Bhattacharyya, A.K. Verma, M.S. Mandal, A.J. Chamkha, D. Yadav, Unsteady stagnation-point flow of CNTs suspended nanofluid on a shrinking/expanding sheet with partial slip: multiple solutions and stability analysis, Waves Random Complex Media 35 (3) (2025) 4593–4614.
[40] N. Abbas, W. Shatanawi, T.A.M. Shatnawi, Thermodynamic feature of PHF and PST with Sisko fluid flow over a Riga curved stretching surface, Mod. Phys. Lett. B 39 (10) (2025) 2450456.
[41] K. Vafai, C.L. Tien, Boundary and inertia effects on flow and heat transfer in porous media, Int. J. Heat. Mass Transf. 24 (2) (1981) 195–203.
[42] S. Jayanthi, M. Kumari, Effect of variable viscosity on non-Darcy free or mixed convection flow on a vertical surface in a non-Newtonian fluid saturated porous medium, Appl. Math. Comput. 186 (2) (2007) 1643–1659.
[43] A. Aziz, Y. Ali, T. Aziz, J.I Siddique, Heat transfer analysis for stationary boundary layer slip flow of a power-law fluid in a Darcy porous medium with plate suction/injection, PLoS. One 10 (9) (2015) e0138855.
[44] T.A.M. Shatnawi, N. Abbas, W. Shatanawi, Comparative study of Casson hybrid nanofluid models with induced magnetic radiative flow over a vertical permeable exponentially stretching sheet, AIMS. Math. 7 (12) (2022) 20545–20564.
[45] S. Ghosh, S. Mukhopadhyay, Flow and heat transfer of nanofluid over an exponentially shrinking porous sheet with heat and mass fluxes, Propuls. Power Res. 7 (3) (2018) 268–275.
[46] M.R. Khan, M.A. Elkotb, R.T. Matoog, N.A. Alshehri, M.A. Abdelmohimen, Thermal features and heat transfer enhancement of a casson fluid across a porous stretching/shrinking sheet: analysis of dual solutions, Case Stud. Therm. Eng. 28 (2021) 101594.
[47] A.K. Verma, K. Bhattacharyya, S. Rajput, M.S. Mandal, A.J. Chamkha, D. Yadav, Buoyancy driven non-Newtonian Prandtl-Eyring nanofluid flow in Darcy-Forchheimer porous medium over inclined non-linear expanding sheet with double stratification, Waves Random Complex Media 35 (3) (2025) 4424–4456.
[48] M. Abd El-Aziz, Dual solutions in hydromagnetic stagnation point flow and heat transfer towards a stretching/shrinking sheet with non-uniform heat source/sink and variable surface heat flux, J. Egypt. Math. Soc. 24 (3) (2016) 479–486.
[49] D. Mythili, R. Sivaraj, Influence of higher order chemical reaction and non-uniform heat source/sink on Casson fluid flow over a vertical cone and flat plate, J. Mol. Liq. 216 (2016) 466–475.
[50] B. Ramandevi, J.R. Reddy, V. Sugunamma, N. Sandeep, Combined influence of viscous dissipation and non-uniform heat source/sink on MHD non-Newtonian fluid flow with Cattaneo-Christov heat flux, Alex. Eng. J. 57 (2) (2018) 1009–1018.
[51] F. Alzahrani, M.I. Khan, Applications of Darcy-Forchheimer 3D reactive rotating flow of rate type nanoparticles with non-uniform heat source and sink and activation energy, Chem. Phys. Lett. 783 (2021) 139054.
[52] B. Manvi, J. Tawade, M. Biradar, S. Noeiaghdam, U. Fernandez-Gamiz, V. Govindan, The effects of MHD radiating and non-uniform heat source/sink with heating on the momentum and heat transfer of Eyring-Powell fluid over a stretching, Results. Eng. 14 (2022) 100435.
[53] N. Abbas, K.U. Rehman, W. Shatanawi, K. Abodayeh, Mathematical model of temperature-dependent flow of power-law nanofluid over a variable stretching Riga sheet, Waves Random Complex Media 35 (3) (2025) 10012–10029.
[54] A.K. Verma, S. Rajput, K. Bhattacharyya, A.J. Chamkha, D. Yadav, Comparison between graphene-water and graphene oxide-water nanofluid flows over exponential shrinking sheet in porous medium: dual solutions and stability analysis, Chem. Eng. J. Adv. 12 (2022) 100401.
[55] A.K. Gautam, S. Rajput, K. Bhattacharyya, A.K. Verma, M.G. Arif, A.J. Chamkha, Existence of multiple solutions for magnetohydrodynamic flows of second-grade and Walter's B fluids due continuously contracting flat sheet with partial slip, Partial Differ. Equ. Appl. Math. 6 (2022) 100434.
[56] P.D. Weidman, D.G. Kubitschek, A.M. Davis, The effect of transpiration on self-similar boundary layer flow over moving surfaces, Int. J. Eng. Sci. 44 (11–12) (2006) 730–737.
[57] S.D. Harris, D.B. Ingham, I. Pop, Mixed convection boundary-layer flow near the stagnation point on a vertical surface in a porous medium: brinkman model with slip, Transp. Porous. Media 77 (2) (2009) 267–285.
[58] S. Parvin, S.S.P.M. Isa, F.S. Al-Duais, S.M. Hussain, W. Jamshed, R. Safdar, M.R. Eid, The flow, thermal and mass properties of Soret- Dufour model of magnetized Maxwell nanofluid flow over a shrinkage inclined surface, PLoS. One 17 (4) (2022) e0267148.
[59] N.A. Zainal, R. Nazar, K. Naganthran, I. Pop, Unsteady flow of a Maxwell hybrid nanofluid past a stretching/shrinking surface with thermal radiation effect, Appl. Math. Mech. (Engl. Ed.) 42 (10) (2021) 1511–1524.

UNCORRECTED PROOF